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1. INTRODUCTION

Gap: Driver facial expression recognition, including DFER-GCViT (Saadi et al., 2023) at 98.27% on KMU-FED [1], has never been verified under real-world occlusions [1].

Why it matters: Grad-CAM confirms heavy reliance on the mouth region [1], which is vulnerable to masks and sunglasses.

This study: synthetic occlusions applied to KMU-FED [3]. Robustness measured via accuracy drop (%).

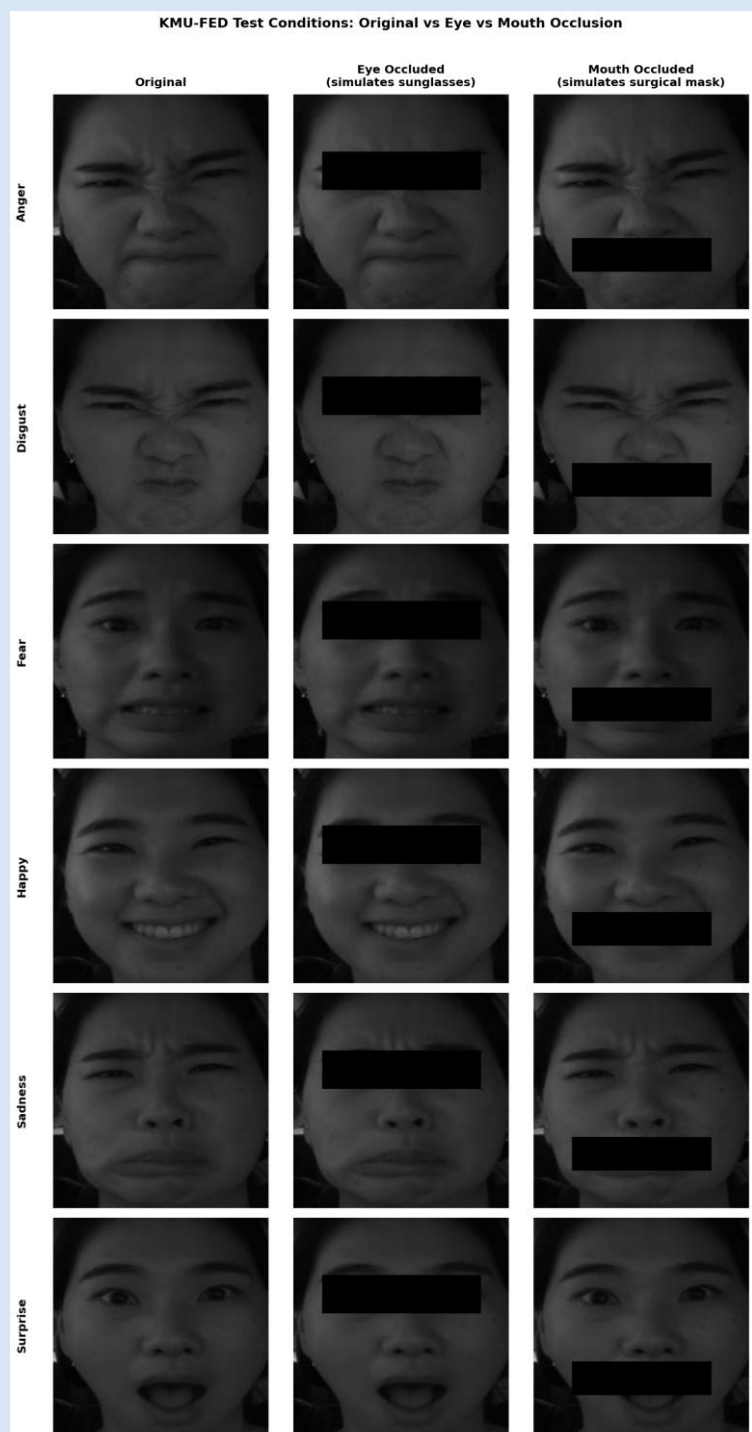


Fig. 1. Mouth occlusion erases the signal the baseline model depends on, the root cause of its 60% accuracy drop.

2. RESEARCH QUESTIONS

RQ1: How much does DFER-GCViT accuracy and F1-score drop under synthetic occlusion at test time?

RQ2: Can training with occluded samples reduce that performance gap?

3. METHODOLOGY

Step 1, Baseline (Model A): pretrained GC-ViT Base [2] with the modified classifier head from Saadi et al. [1], evaluated with 10-fold CV on KMU-FED [3].

Step 2, Occlusion: black rectangles placed over the eye (simulating sunglasses) and mouth (simulating a surgical mask), applied programmatically with no manual labelling.

Step 3, Training: Model A trains on clean images only. Model B trains on 50% occluded images (25% eye, 25% mouth), using the same held-out folds.

Step 4, Evaluation: both models (A and B) are tested on three conditions, clean, eye-occluded, and mouth-occluded, using accuracy drop (%) as the robustness metric.

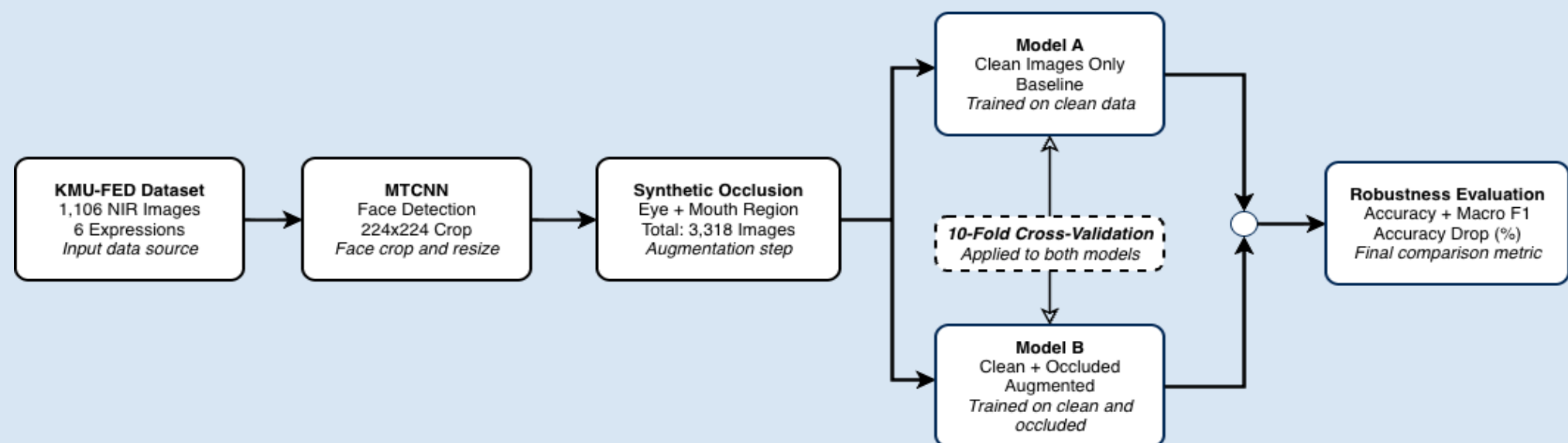


Fig. 2. Testing eye and mouth occlusion separately reveals exactly which region the model relies on.

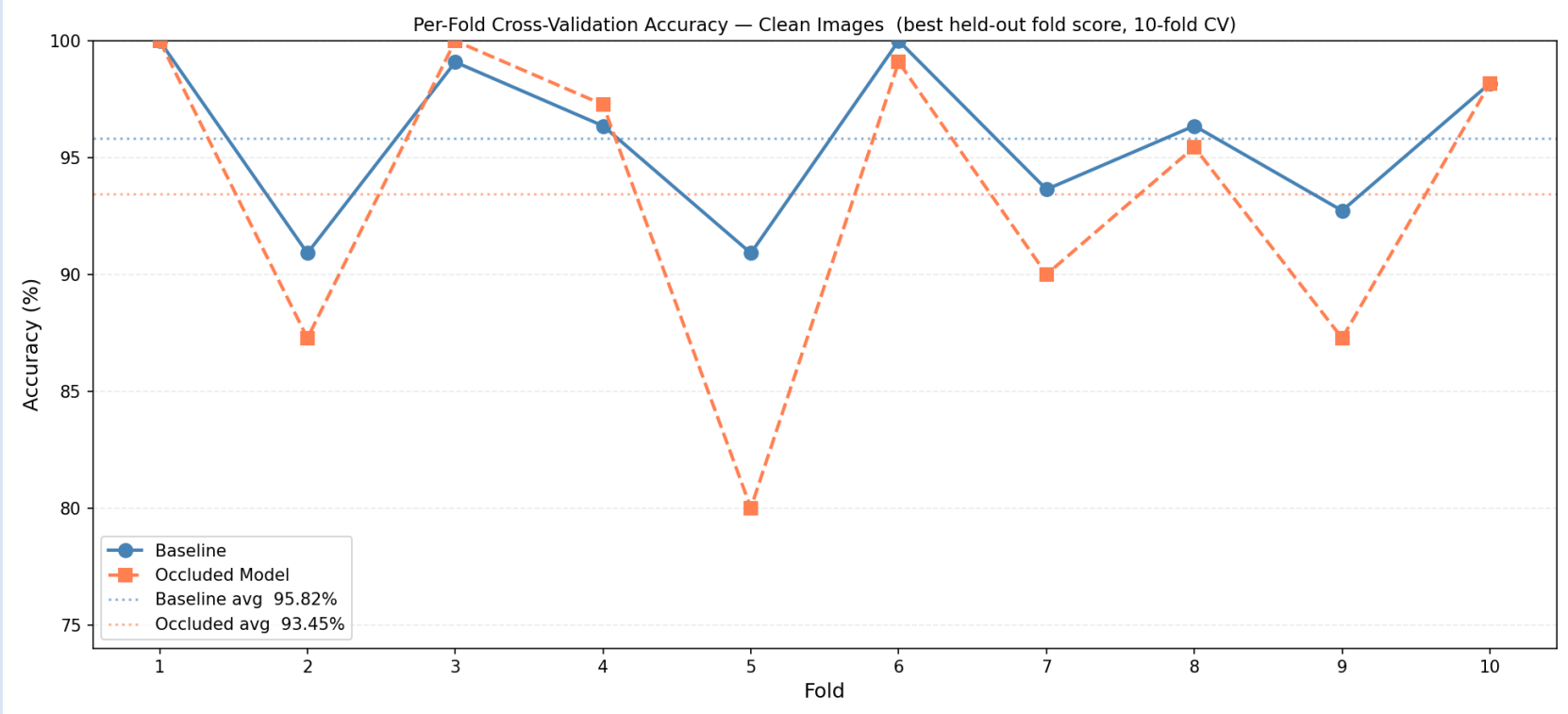


Fig. 3. Per-fold cross-validation accuracy on clean images. Both models perform comparably baseline averages 95.82%, occluded model 93.45% - though the occluded model is less stable, dropping to 80% in fold 5.

4. RESULTS

Test Condition	Model A (Baseline)	Model B (Occluded)
Clean	95.82%	93.45%
Eye Occluded	92.64%	90.73%
Mouth Occluded	35.45%	74.73%
Eye Drop	3.18%	2.73%
Mouth Drop	60.36%	18.73%

Model A relies almost entirely on the mouth: 60.36% accuracy drop vs. just 3.18% for eye occlusion.

Model B recovers most of that loss, cutting the mouth-occlusion drop from 60.36% to 18.73%.

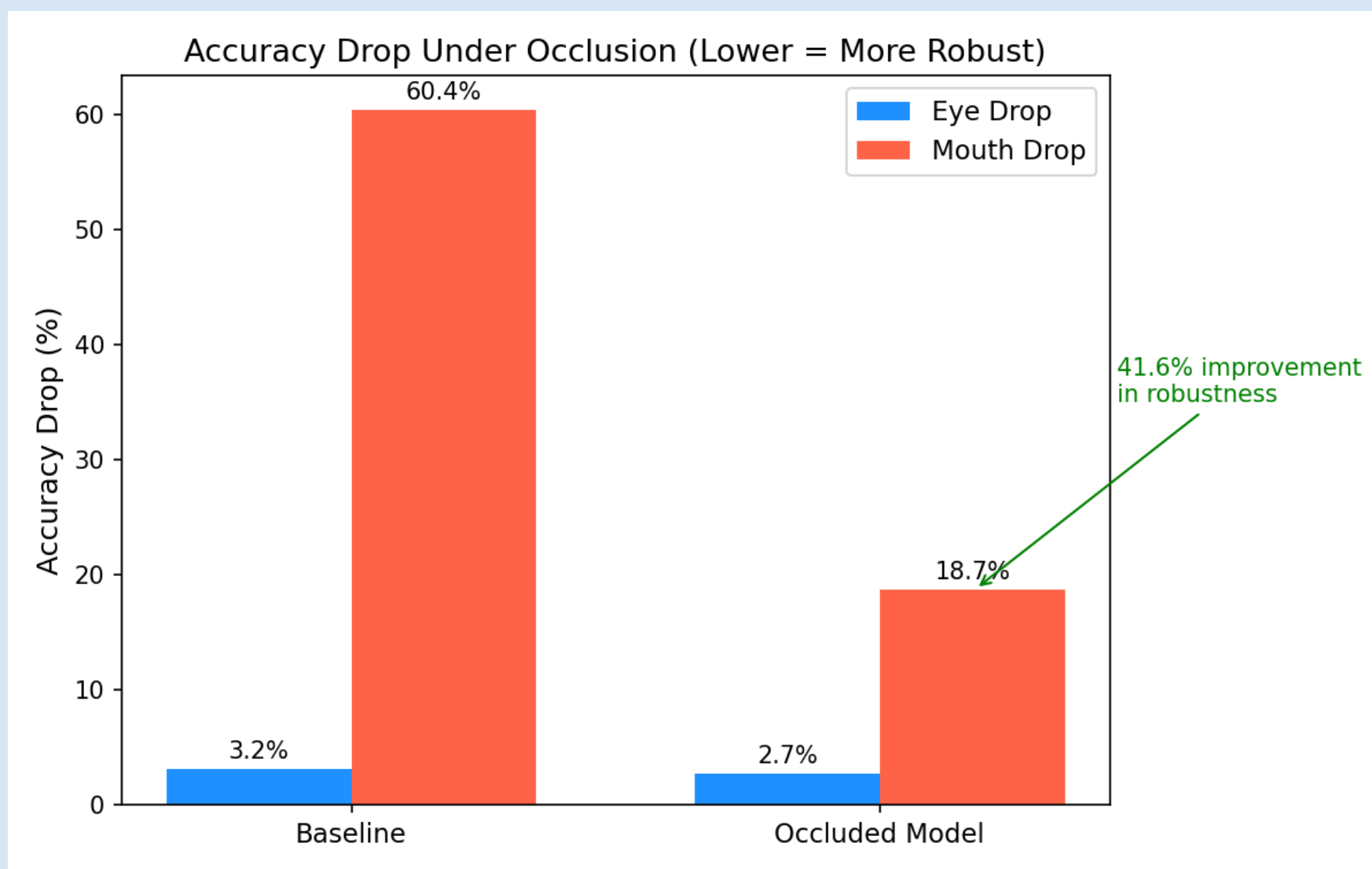


Fig. 4. Accuracy drop (%) under occlusion. Mouth occlusion drops Model A's accuracy by 60.36 percentage points, reduced to 18.73 with occlusion training, a 41.6-point recovery. Eye occlusion has minimal effect on both models.

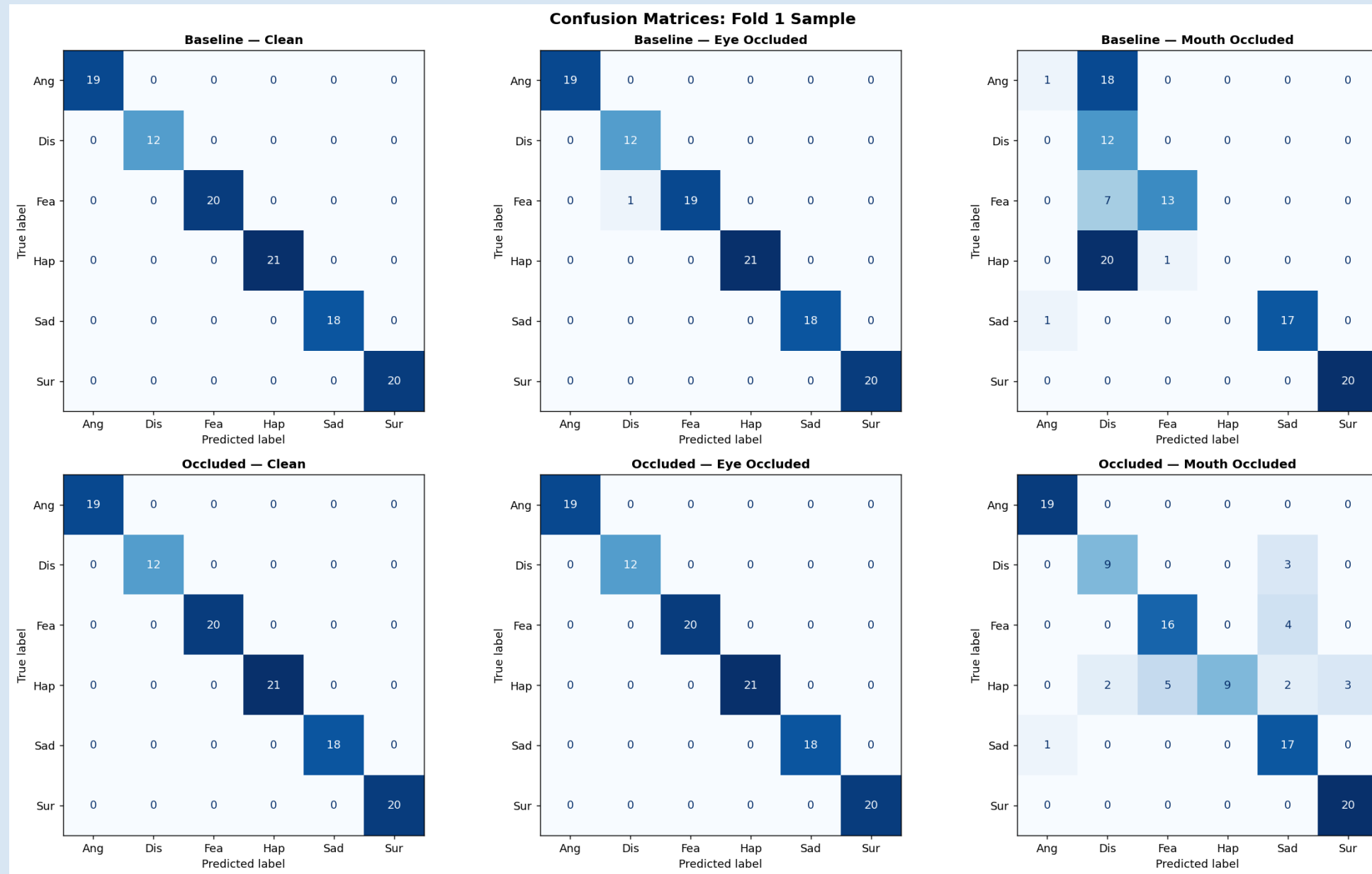


Fig. 5. Confusion matrices under mouth occlusion. Model A (baseline) confuses Anger and Happy with Disgust; Model B (occluded training) stays mostly on the diagonal.

5. CONCLUSION

Mouth occlusion is catastrophic for **Model A**: expressions driven by mouth signals (Happy, Surprise, Anger) fail completely.

Model B cuts the mouth-occlusion drop from 60.36% to 18.73%, while clean accuracy drops slightly from 95.82% to 93.45%.

GC-ViT global attention makes both models resilient to eye occlusion. Mouth remains the critical vulnerability.

6. FUTURE WORK

- Test with real occlusion data, using subjects wearing actual sunglasses and surgical masks during recording.
- Multi-modal fusion combining facial expression recognition with audio and physiological signals.
- Real-time deployment on embedded in-vehicle systems.

References:

- [1] I. Saadi, D. Cunningham, A. Taleb-Ahmed, A. Hadid, Y. El Hillali, "Driver's Facial Expression Recognition using Global Context Vision Transformer," IEEE CVMI, 2023.
- [2] A. Hatamizadeh, H. Yin, J. Kautz, P. Molchanov, "Global Context Vision Transformers," arXiv:2206.09959, 2022.
- [3] M. Jeong, B. C. Ko, "Driver's facial expression recognition in real-time for safe driving," Sensors, vol. 18, no. 12, p. 4270, 2018.